

Yoga Pose classification

[SeleeMlcker/Yoga_Pose_Classification: Real-time yoga pose classification using a webcam, MediaPipe, and Trained Keras Model. This project is using MediaPipe extract the Yoga pose landmark \(33 Body Landmark\) in XYZ values to train the Model. It enables the model to classify 5 different Yoga pose in real-time.](#)

Overview

Real-time yoga pose classification using webcam, MediaPipe and trained Keras Model in google colab.

This project is using MediaPipe extract the Yoga pose landmark (33 Body Landmark) in XYZ values to train the model. It enables the model to classify User's Yoga Pose in real time.

The project goes from Google colab for model training to virtual machines for real-time detection.

Model training

The first time for this project is to train a model able to classify different yoga poses.

Dataset

The dataset used for training is from Kaggle, Yoga Pose Dataset by Niharika41298, source: [Yoga Poses Dataset](#).

The dataset contains images of 5 yoga poses:

- Downdog (196 images)
- Goddess (199 images)
- Plank (197 images)
- Tree (198 images)
- Warrior2 (198 images)

Images were used to extract MediaPipe pose landmarks for training the classifier.

Google Colab

Google Colab is a free, browser-based coding environment where users can write and run Python code without installing anything.

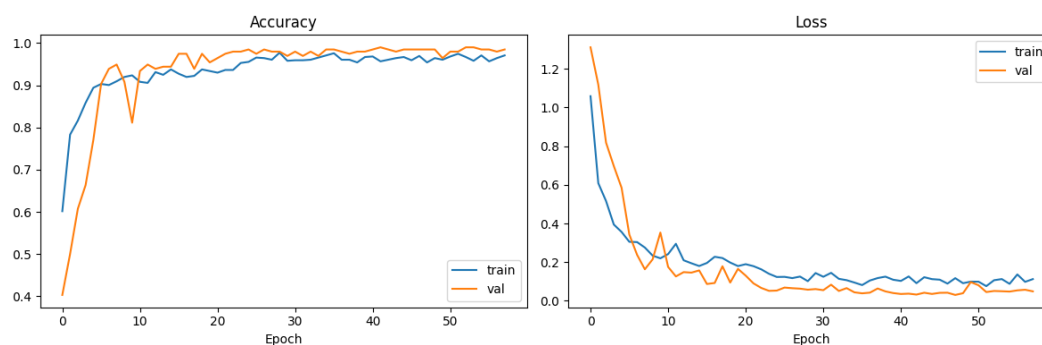
We used Google Colab to preprocess images and model training.

Source: [yoga_pose_estimation.ipynb - Colab](#)

The first step to preprocess images is using Mediapipe to detect the body landmark from the yoga pose and store the data in csv file.

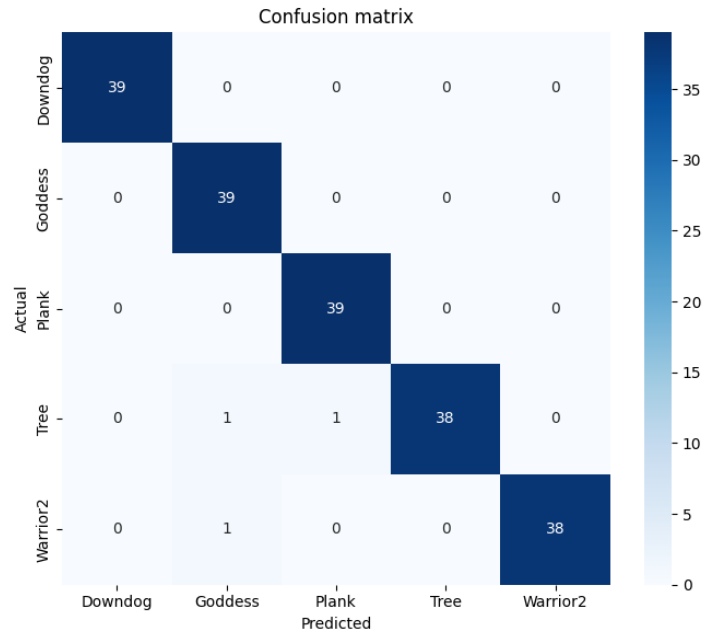
	A	B	C	D	E	F	G	H	I
1	label	kp0_x	kp0_y	kp0_z	kp0_vis	kp1_x	kp1_y	kp1_z	kp1_vis
2	Downdog	0.4361152053	0.7043814659	-0.01018138602	0.9996632338	0.4154409766	0.7072657347	-0.04278697073	0.999722085
3	Downdog	0.5093089342	0.7265041471	-0.05422639474	0.9992573857	0.4923278391	0.7477490902	-0.09305482358	0.999582291
4	Downdog	0.3922438025	0.6819796562	-0.08610557765	0.9999011755	0.373059392	0.686172843	-0.1341161281	0.9999328852
5	Downdog	0.5531132817	0.7668201923	-0.03063597158	0.9999866486	0.5741819739	0.7749217749	-0.01444951445	0.9999835491
6	Downdog	0.5426728129	0.7607399225	-0.09094608575	0.9999136925	0.5640704632	0.7715965509	-0.0712518692	0.9998960495
7	Downdog	0.4030862451	0.6476160288	-0.09086273611	0.9952170849	0.3855918944	0.6520059109	-0.1229212955	0.9969519377
8	Downdog	0.497772038	0.6825879812	-0.1152154654	0.9999083281	0.5140674114	0.6998786926	-0.09286426753	0.999866724
9	Downdog	0.6625438929	0.7466964126	-0.0747255981	0.9994927645	0.6807221174	0.7447912097	-0.05139759555	0.9995187521
10	Downdog	0.36610654	0.7396700978	-0.0967367813	0.9952688813	0.346177578	0.7285676599	-0.1406515539	0.9959713817
11	Downdog	0.6625438929	0.7466964126	-0.0747255981	0.9994927645	0.6807221174	0.7447912097	-0.05139759555	0.9995187521
12	Downdog	0.4452614188	0.7296004295	-0.0632121712	0.9996863604	0.4263928533	0.749984026	-0.08545117825	0.9997985959
13	Downdog	0.4452614188	0.7296004295	-0.0632121712	0.9996863604	0.4263928533	0.749984026	-0.08545117825	0.9997985959
14	Downdog	0.3700273633	0.5338650346	0.0709817782	0.9998310804	0.352339983	0.5481821299	0.04507424682	0.9998742342
15	Downdog	0.3951483071	0.7231779099	-0.09400351346	0.9870623946	0.3805373609	0.7232505083	-0.1198532358	0.9919714928
16	Downdog	0.4271848798	0.6596130133	-0.04729207978	0.9989037514	0.411362052	0.6777103543	-0.08351682872	0.9993678543

Now, we can use the number to train the model by finding the patterns and differences between each pose.



Accuracy -- "How often was the model right?"

Loss -- "How wrong was the mode, and by how much?"



The “1” means the model saw a real Tree pose but called Warrior2.

	precision	recall	f1-score	support
Downdog	1.00	1.00	1.00	39
Goddess	0.95	1.00	0.97	39
Plank	0.97	1.00	0.99	39
Tree	1.00	0.95	0.97	40
Warrior2	1.00	0.97	0.99	39
accuracy			0.98	196
macro avg	0.99	0.98	0.98	196
weighted avg	0.99	0.98	0.98	196

A classification report tells how well the model predicted each yoga pose on the test set. In overall, 98% accuracy across 196 test images. The only slight weaknesses are Goddess and Tree occasionally getting confused with each other.

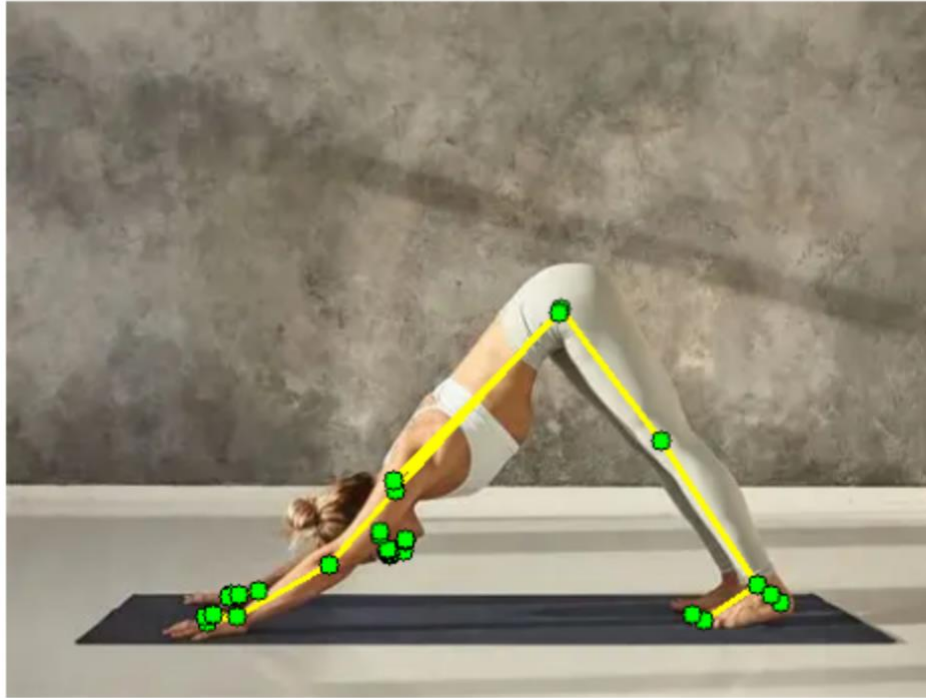
Demonstration

Top 3 predictions:

Downdog	100.0%
Plank	0.0%
Warrior2	0.0%



Predicted: Downdog (100.0%)



The model is able to detect the yoga post by pictures, videos, and real-time.